QuietDesk: Components List

Meta Global AI Developer Hackathon · Raspberry Pi 5 voice-first AI companion · Shopping checklist

1. Main components (zaroori)

	#	Component	Qty	Andaaze ka price
☐	1	**Raspberry Pi 5** (8GB best, 4GB bhi chalega) Device ka dimaag	1	₹7,000 – 10,000
☐	2	**Pi 5 Active Cooler** (official) Pi 5 garam hota hai, zaroor lagao	1	₹500 – 700
☐	3	**microSD card** 32 ya 64GB, A2 class (SanDisk / Samsung) OS aur code ispe chalega	1	₹500 – 800
☐	4	**Official 27W USB-C power supply** Development ke time table pe use karne ke liye	1	₹1,200 – 1,500
☐	5	**Waveshare 2.9" e-Paper module** (SPI, wire cable ke saath) Bare module lena, HAT nahi (GPIO18 clash). Budget kam ho toh 2.13" le lo.	1	₹1,500 – 2,500
☐	6	**MAX98357A I2S amplifier** Pi 5 mein audio jack nahi hai, sound isi se niklegi	1	₹250 – 400
☐	7	**Speaker 4Ω ya 8Ω, 3W** (40–50mm) Amp ke screw terminal se judega	1	₹100 – 250
☐	8	**USB microphone** (mini plug-in) Seedha USB port mein, wiring nahi chahiye	1	₹300 – 700
☐	9	**Tactile push button** 12mm Push-to-talk. Extra rakho.	3 – 4	₹10 – 20 each
☐	10	**LED 5mm** Status light	3 – 4	₹5 each
☐	11	**330Ω resistor** LED ke saath series mein	5	₹10 – 20
☐	12	**Breadboard** 400-point (half size) Button, LED aur GND rail ke liye	1	₹80 – 150
☐	13	**Jumper wires**: F-F, M-F, M-M (40 each) E-ink aur amp ke liye F-F, breadboard ke liye M-F	1 set each	₹150 – 250

2. Portable banane ke liye

	#	Component	Qty	Andaaze ka price
☐	14	**USB-C PD power bank** (10,000 – 20,000mAh) Demo ke time battery pe chalane ke liye. 5V 3A se kaam chalega, bas Pi "low power" warning dega.	1	₹1,500 – 2,500

3. Tools (agar pehle se nahi hain)

	#	Item	Kyun chahiye
☐	15	**Soldering iron + solder wire**	MAX98357A aksar header pins **bina solder** ke aata hai. Kai e-ink modules ke saath bhi aisa hota hai.
☐	16	**microSD card reader**	Laptop se Raspberry Pi OS flash karne ke liye
☐	17	**Multimeter (optional)**	Wiring aur voltage check karne ke liye
☐	18	**Micro-HDMI to HDMI cable (optional)**	Screen pe setup karna ho toh. Warna SSH se headless setup ho jayega.

4. Pehle se hona chahiye

- **Android phone**: companion app ke liye (notifications relay)
- **Laptop**: coding aur Pi setup ke liye
- **Wi-Fi / hotspot**: Pi aur phone dono **ek hi network** pe hone chahiye

5. Total kharcha (andaaza)

Pi ke saath
₹13,000 – 18,000

Agar Pi pehle se hai
₹4,000 – 7,000

Prices andaaze se hain (Robu.in, Robocraze, Amazon India). Order se pehle check kar lena.

6. Wiring quick reference (physical pin numbers)

Part	Part pin	Pi 5 pin	Part	Part pin	Pi 5 pin
E-ink	VCC	1 (3V3)	Amp	VIN	2 (5V)
E-ink	GND	9 (GND)	Amp	GND	14 (GND)
E-ink	DIN	19 (GPIO10)	Amp	BCLK	12 (GPIO18)
E-ink	CLK	23 (GPIO11)	Amp	LRC	35 (GPIO19)
E-ink	CS	24 (GPIO8)	Amp	DIN	40 (GPIO21)
E-ink	DC	22 (GPIO25)	Button	Leg 1 / Leg 2	13 (GPIO27) / GND rail
E-ink	RST	11 (GPIO17)	LED	330Ω → LED +	15 (GPIO22)
E-ink	BUSY	18 (GPIO24)	LED / Breadboard	LED − / GND rail	GND rail ← 25 (GND)

Zaroori tips:

- **Aaj hi order karo.** Deadline Nov 4 (9 PM ET) hai, aur delivery mein 2–4 din lag sakte hain.
- **Pi 5 hi lena, Pi 4 nahi.** Wiring aur config Pi 5 ke hisaab se hain.
- E-ink **SPI module** ho aur **cable saath aaye**, ye check karna.
- Pin 1 SD card slot ke sabse paas wala pin hai. Config mein: `dtparam=spi=on` aur `dtoverlay=max98357a`.